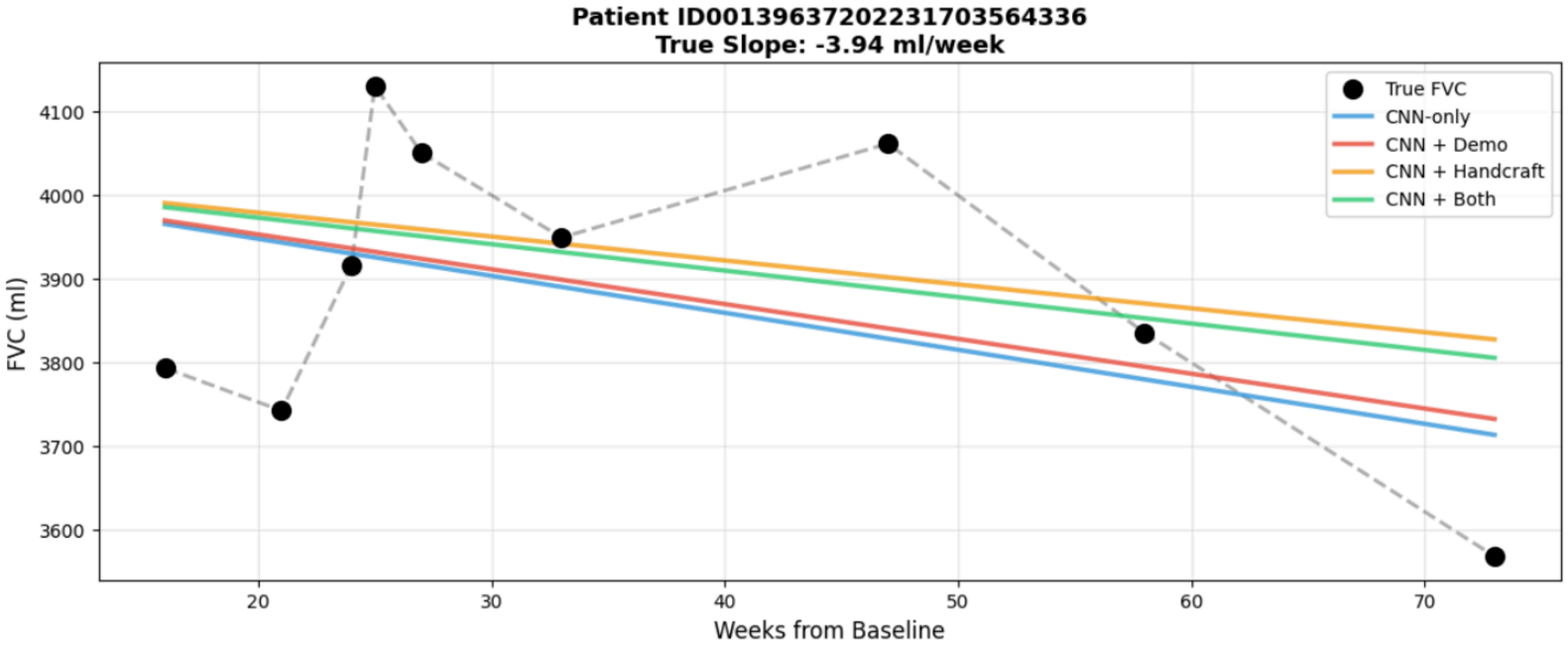
PROGRESS PREDICTION

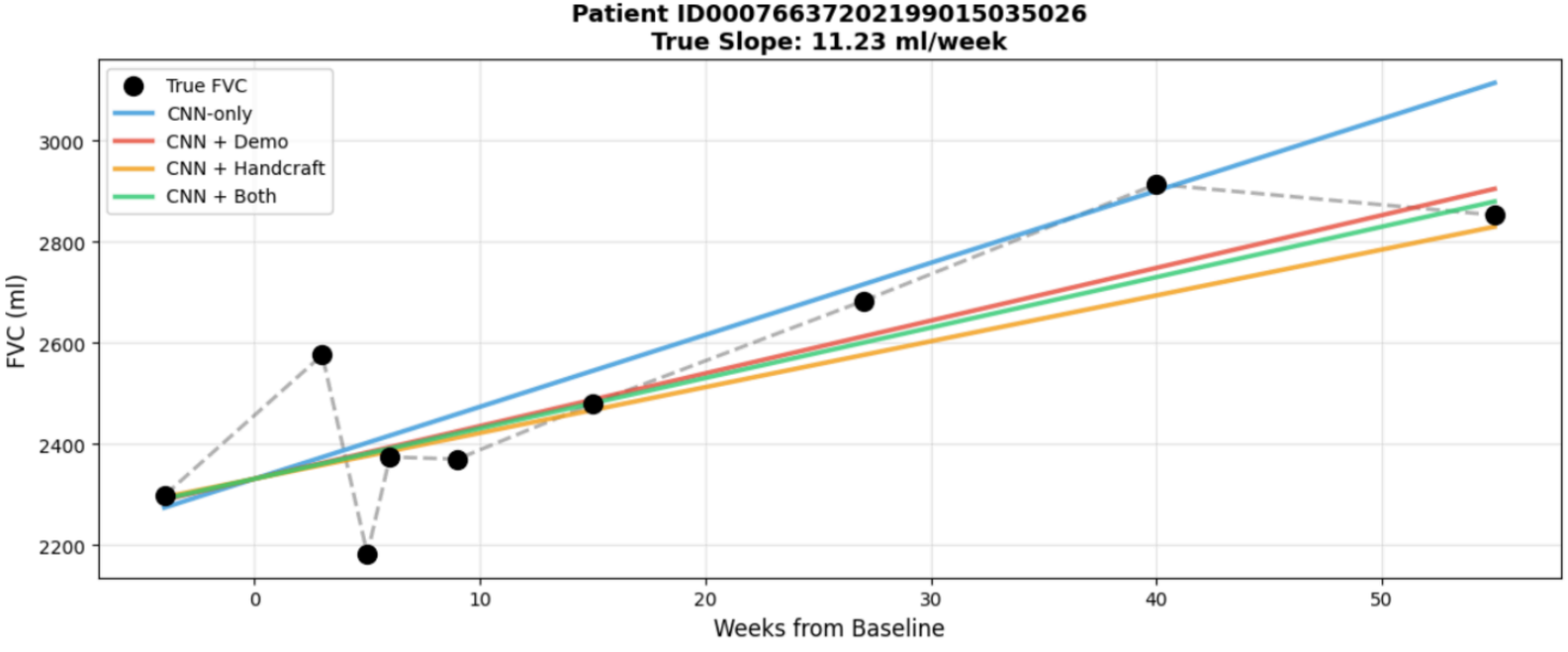
Model	FVC MAE (ml)
Linear Regression	564.42
Polynomial 2	718.73
Polynomial 3	794.53
CNN – Polynomial 3	682.73
CNN – Slope	128.38
CNN - Slope + Handcraft features + Demographics	**106.48**

CNN - Slope + Handcrafted features + Demographics	**106.48**

Performance divided between 3 sets:

- CNN - Slope only : 128.38 ml
- CNN + Demographics : 115.00 ml
- CNN + Handcrafted : 106.86 ml
- CNN + Demographics + Handcrafted : **106.48 ml**


Patient ID00139637202231703564336
True Slope: -3.94 ml/week
True FVC
CNN-only
CNN + Demo
CNN + Handcraft
CNN + Both
FVC (ml)
4100
4000
3900
3800
3700
3600
20
30
40
50
60
70
Weeks from Baseline


Patient ID00076637202199015035026
True Slope: 11.23 ml/week
True FVC
CNN-only
CNN + Demo
CNN + Handcraft
CNN + Both
FVC (ml)
3000
2800
2600
2400
2200
0
10
20
30
40
50
Weeks from Baseline

Approaches adopted for progression prediction :

1. Define FVC from slope generated by CNN+Handcrafted+Demographics and check progression

2. Predict FVC @ week 52 from Feature Embedding - Handcrafted - demographics and check progression

3. Predict progression risk directly from Feature Embedding - Handrafted - Demographics

1. Define FVC from slope generated by CNN+Handcrafted+Demographics and check progression

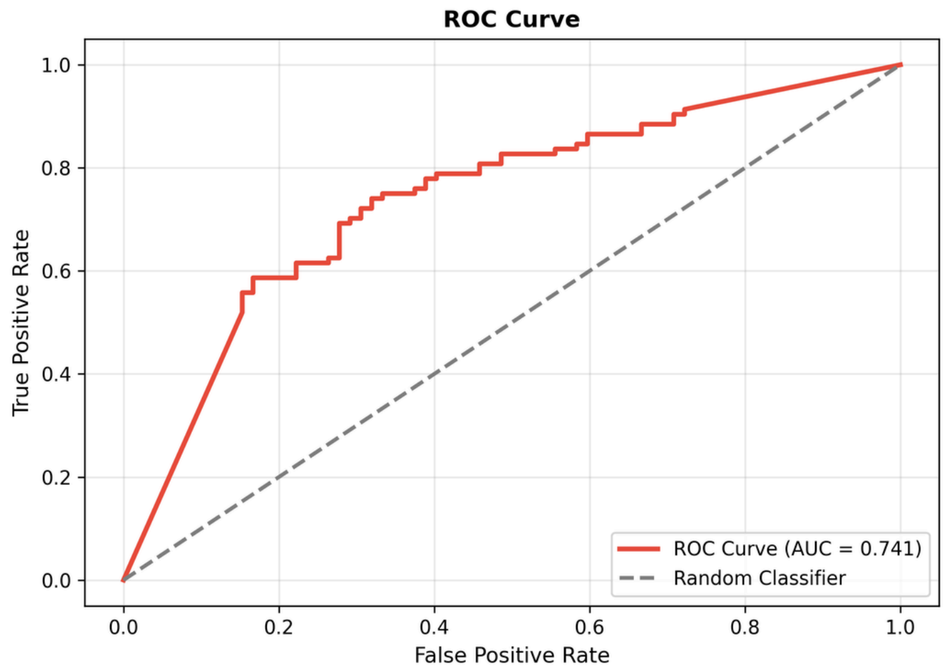
How labels defined:

- If ANY future measurement shows ⩾10% decline → Label = 1 (Progression)
- If ALL future measurements are <10% decline → Label = 0 (Stable)

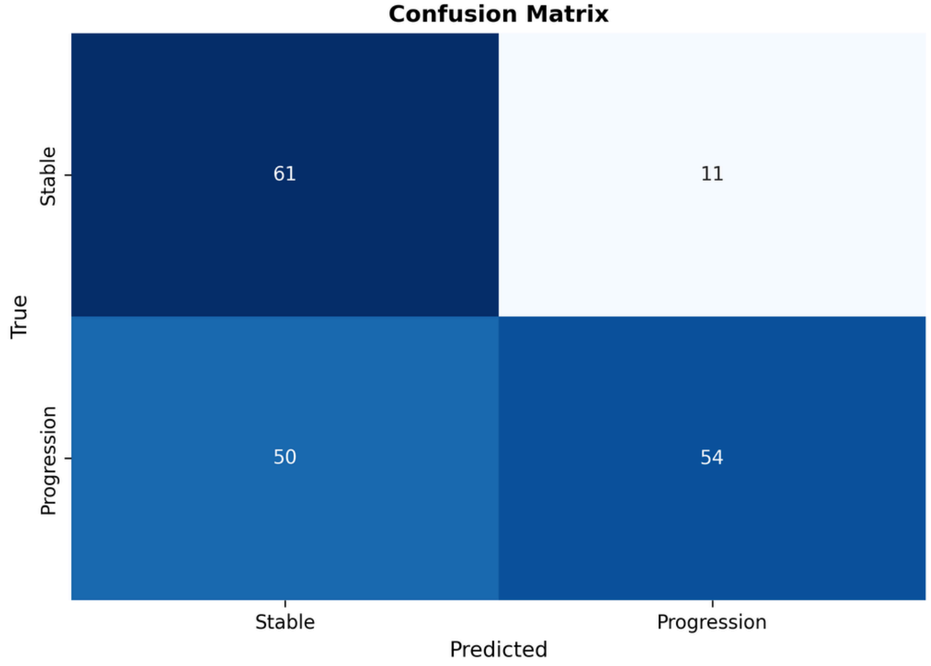
Steps:

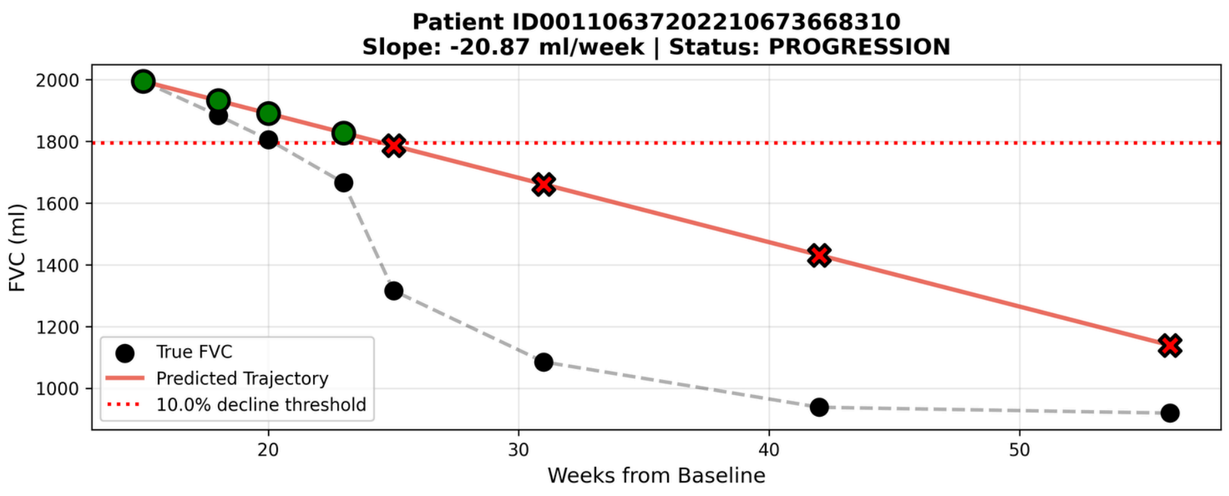
- Compute FVC(52) → FVC(t) = slope * week(t) + intercept at week 0
- FVC(52) - intercept at week 0
- Compare labels

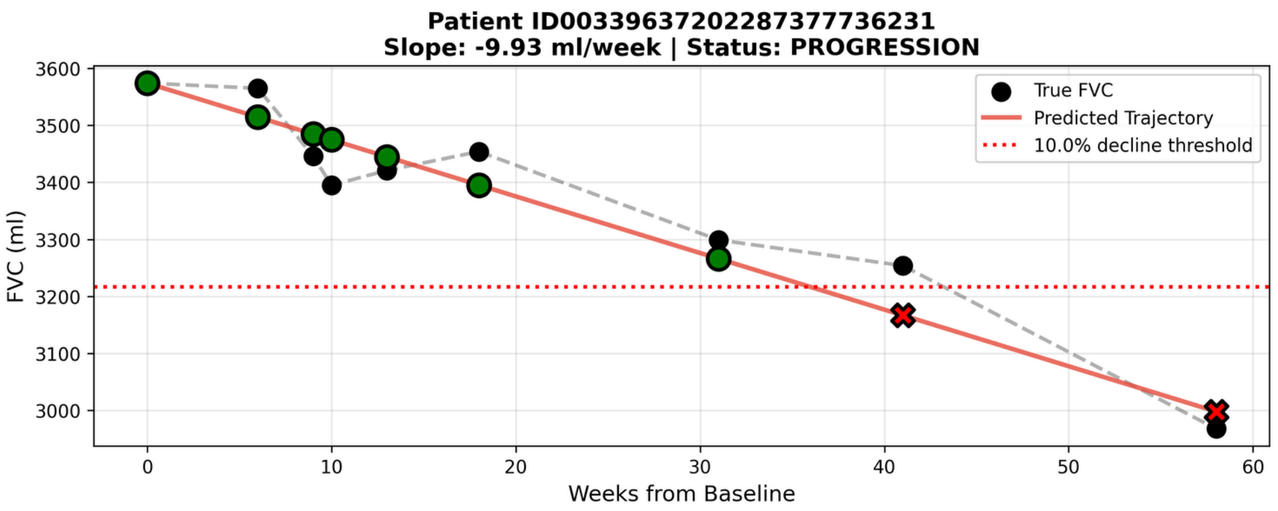
Accuracy : 0.653 ± 0.027
Precision : 0.831 ± 0.066
Recall : 0.519 ± 0.010
Specificity : 0.847

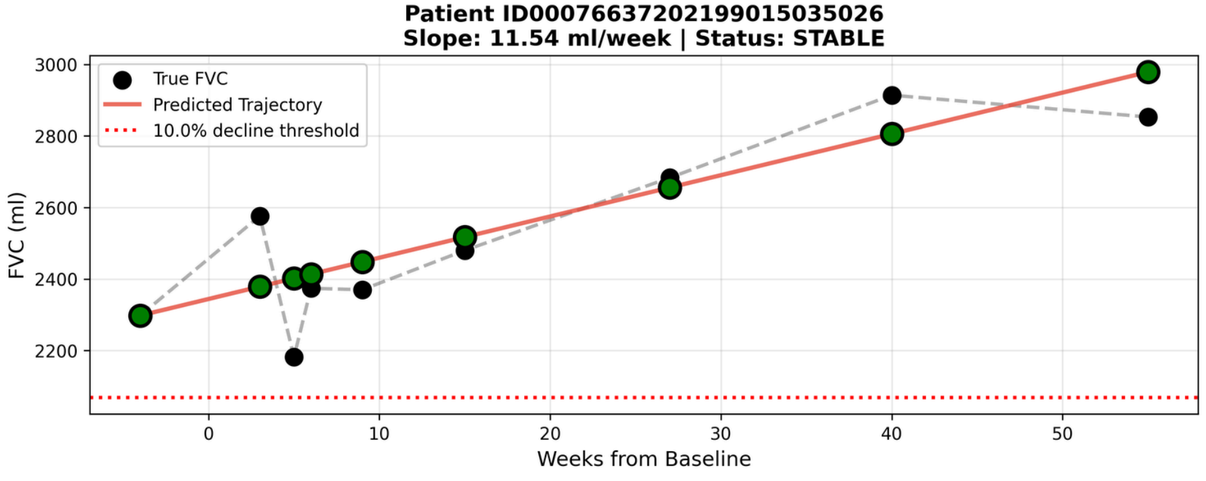


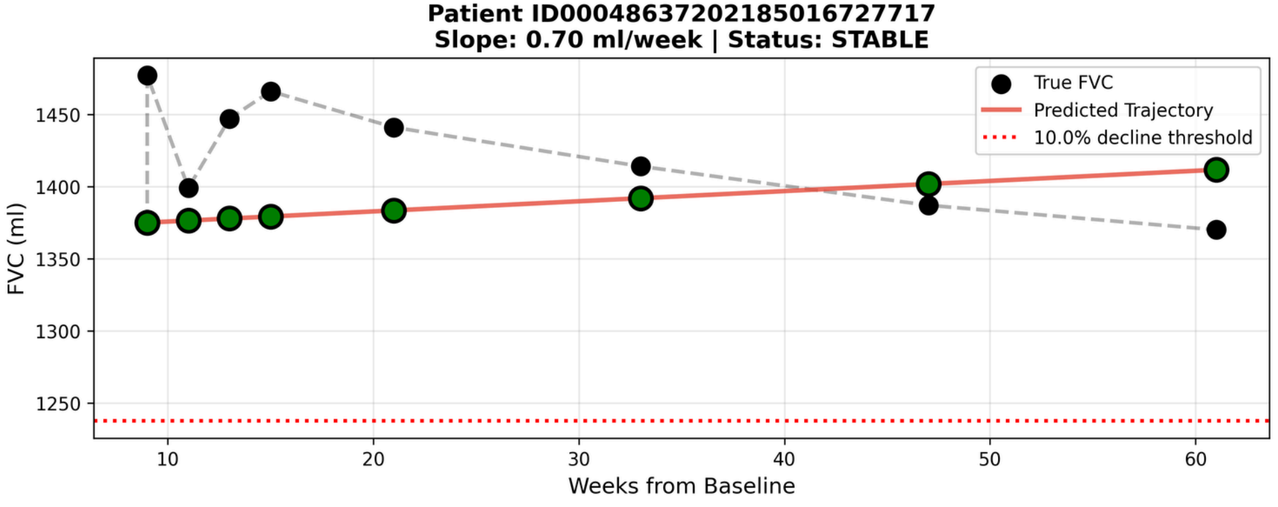
Problem:
Model is conservative, avoids false alarms, but misses some true progressions.
We don't want to miss progressions as it is worse than a false alarm.
Need threshold tuning (tolerance) as many patients lie on a 9% threshold.




Patient ID00110637202210673668310
Slope: -20.87 ml/week | Status: PROGRESSION
FVC (ml)
Weeks from Baseline
True FVC
Predicted Trajectory
10.0% decline threshold


Patient ID00339637202287377736231
Slope: -9.93 ml/week | Status: PROGRESSION
FVC (ml)
Weeks from Baseline
True FVC
Predicted Trajectory
10.0% decline threshold


Patient ID00076637202199015035026
Slope: 11.54 ml/week | Status: STABLE
FVC (ml)
Weeks from Baseline
True FVC
Predicted Trajectory
10.0% decline threshold


Patient ID00048637202185016727717
Slope: 0.70 ml/week | Status: STABLE
FVC (ml)
Weeks from Baseline
True FVC
Predicted Trajectory
10.0% decline threshold

2. Predict FVC @ week 52 from Feature Embedding - Handcrafted - demographics and check progression

Used Feature Embeddings extracted from CNN added to Handcrafted and Demographics

- MLP : concatenate the feature embedding to the rest and predict FVC

Used a 5 Kfold Cross validation:

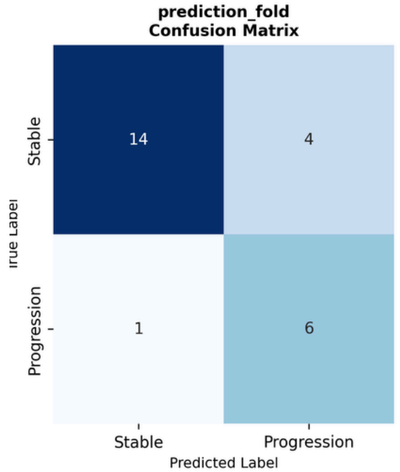
MAE (FVC(52)) : **571.64 ± 142.75 mL**

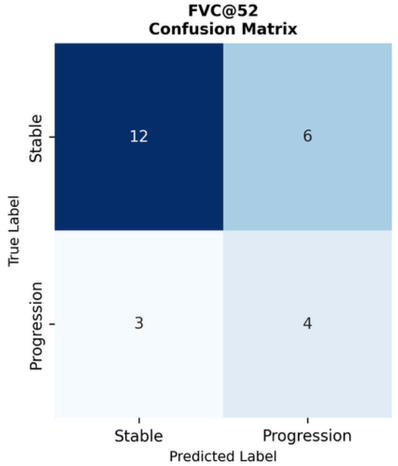
High variance between each fold.
Difficult to predict the FVC from these starting features.

Computes Progression prediction as done for the slope but here just for the FVC values at 52 week as output of the model, therefore restricted dataset (25 patients).

Comparison for those 25 patients between 1st Approach and 2nd Approach:

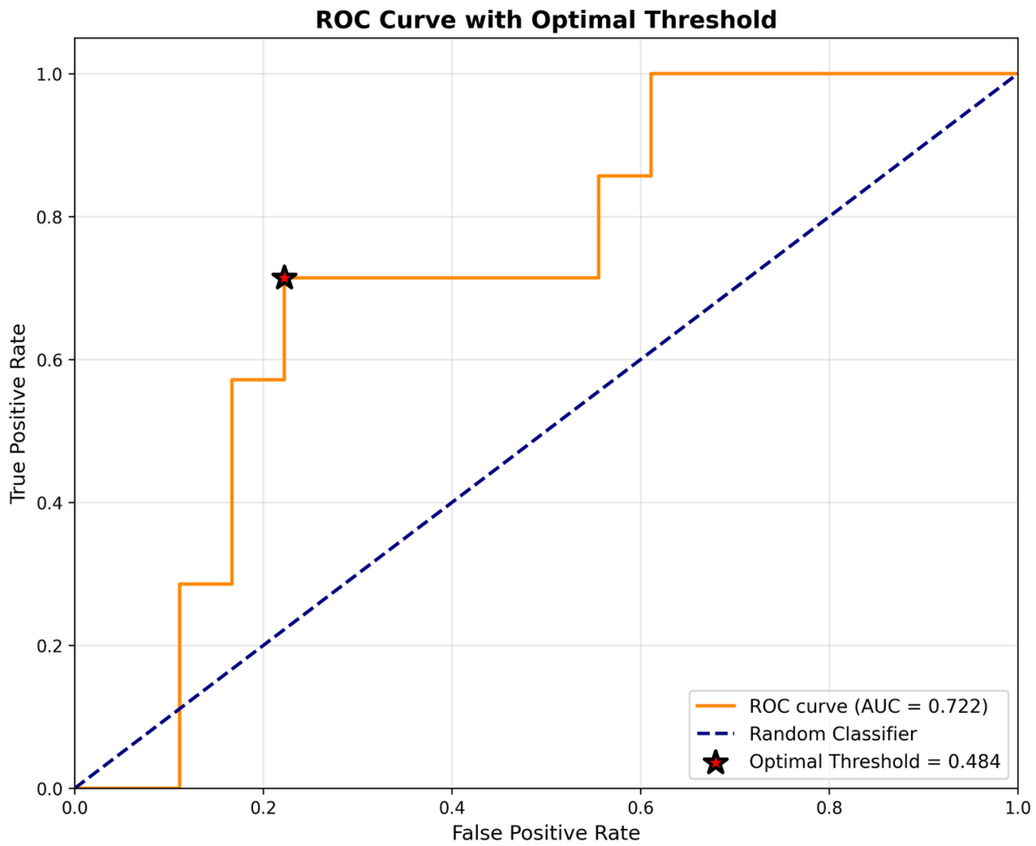
Model	**Accuracy**	**Precision**	**Recall**	**Specificity**	**AUC**
Slope	**0.800**	**0.600**	**0.857**	**0.778**	**0.833**
FVC@52	0.640	0.400	0.571	0.667	0.722





For the restricted dataset the 1st Approach shows better results than the 2nd Approach.

3. Predict progression risk directly from Feature Embedding - Handrafted - Demographics

Use Feature Embeddings + Handcrafted features + Demographics to directly predict progression → output is a probability risk

Accuracy: 0.7600
Precision: 0.5556
Recall: 0.7143